Linguistic Features of German Easy and Plain Language: A Cross-Dataset Analysis

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Abstract

This paper investigates linguistic differences between Standard German and two simplification varieties, Plain Language and Easy Language. Using publicly available German datasets, we compute surface-level indicators including token and sentence length, vocabulary-frequency features, readability scores, and selected guideline-related phenomena such as subjunctive forms and complex negation expressions. We treat Plain Language and Easy Language as separate targets and compare their feature profiles with original or reference corpora where available. From a cross-dataset perspective, we ask whether similar patterns recur across independently constructed resources or whether simplified language profiles are dataset-specific. Since the resources are heterogeneous and not uniformly parallel, we interpret the results as descriptive corpus-level comparisons rather than causal effects of simplification.

1 Introduction

Text simplification has become an important direction in Natural Language Processing (Siddharthan, 2014; Madina et al., 2023), as public communication increasingly needs to be understandable for heterogeneous audiences (International Organization for Standardization, 2023). However, adoption (e.g. in the form of web sites providing a simplified version) is still lacking (Asghari et al., 2023).

In the German context, two standardized varieties of simplified language are distinguished, *Leichte Sprache* (Easy Language, EL) and *Einfache Sprache* (Plain Language, PL), following the terminology established by Maaß (2020). While Easy Language targets individuals with cognitive disabilities (Deutsches Institut für Normung, 2025), Plain Language addresses a broader audience with limited literacy or language proficiency (Deutsches Institut für Normung, 2024). Both Plain and Easy Language have their own set of rules, defining what prerequisites must be met for a text to be called Plain or Easy Language (Deutsches Institut für Normung, 2024, 2025). As a result, Easy Language and Plain Language should not be treated as interchangeable labels. Models trained or evaluated on one variety may not generalize to the other, and empirical findings may depend strongly on how a dataset operationalizes simplification.

A central prerequisite for replicable research in this area is the availability of suitable datasets.

Over the last years, several German resources have been published that provide standard-language texts, simplified texts, or both, with varying degrees of parallelism. However, these datasets differ substantially with respect to domain (e.g. administrative vs. encyclopedic content), text type, alignment strategy, and production process (professional editing vs. semi-automatic pipelines). Recent work evaluating LLM-generated text simplifications in German further shows that simplification quality depends strongly on how simplification is operationalized, while such output is generally simpler than the source, it does not always fully meet Easy or Plain Language standards and can introduce content errors (Deilen et al., 2023, 2024). Taken together, these dataset- and production-related differences can introduce systematic variation that may be confounded with the linguistic properties of Easy Language or Plain Language themselves. This concern is supported by Jablotschkin et al. (2024), who find that text genre has a strong effect on the distinctiveness of Easy German from Standard German, suggesting that corpus composition can shape observed linguistic differences independently of the simplification variety itself.

Consequently, it remains an open methodological question to what extent linguistic signatures typically associated with simplification, such as shorter sentences or reduced structural complexity,

are stable across datasets, or instead reflect corpus composition and construction.

This paper addresses this question through a comparative, feature-based analysis of publicly available German simplification datasets. We first compute interpretable linguistic features for each corpus, covering both standard-language texts and simplified texts where available. This allows us to examine corpus-level differences within resources where both original or reference and simplified-language subcorpora are available, while also comparing whether similar patterns can be observed across datasets. Throughout the analysis, we treat Easy Language and Plain Language as separate simplification targets.

Since the resources are not uniformly parallel, we do not interpret observed differences as causal effects of simplification alone. Instead, we use the datasets to compare corpus-level feature profiles associated with Standard, Plain, and Easy Language labels across heterogeneous empirical resources.

Our framework operationalizes linguistic complexity through surface-level measures, such as average token and sentence length, vocabulary-frequency indicators, readability scores, and selected guideline-related phenomena such as subjunctive forms and complex negation expressions. We ask whether simplified-language corpora show the expected reductions in surface-level complexity and guideline-discouraged phenomena, and whether these patterns are consistent across datasets or dataset specific. We make our experiments and code publicly available at `https://github.com/JeaBew/de-simple-features`.

2 Related Work

Existing NLP work on German text simplification predominantly evaluates output quality relative to a reference simplification or predicts a continuous complexity score, rather than checking compliance with specific, codified Easy Language or Plain Language rules.

TextComplexityDE (Naderi et al., 2019) frames complexity as a regression problem based on subjective ratings (complexity, understandability, lexical difficulty) collected from German language learners, without reference to Easy or Plain Language rule sets. The GermEval 2022 Shared Task (Mohtaj et al., 2022) builds directly on TextComplexityDE and likewise treats complexity as a regression problem on a 1–7 scale. EASSE-DE (Stodden, 2024) evaluates the output of simplification systems using standard metrics such as BLEU, SARI, and BERTScore, which assess similarity to a reference text rather than rule conformity. DE-Lite (Jablotschkin et al., 2024) takes a data-driven approach instead, identifying characteristic features of Easy German via n-gram models and Kullback-Leibler divergence against Standard German rather than checking texts against a predefined rule catalog.

However, Easy Language is commonly operationalized via explicit rule catalogs and principles, including guidance grounded in legal and audience-related considerations (Bredel and Maaß, 2016). This is notable because simplification guidelines more broadly have been criticized as vague and under-operationalized (Grabar and Saggion, 2022), in contrast to the codified, rule-based standards that govern German Easy and Plain Language specifically (see Section 3).

Beyond rule-conformity checking, readability and complexity metrics have been used to evaluate simplification quality in adjacent contexts. Brivа-Iglesias and Gil (2025) apply standard readability formulas (Flesch Reading Ease, Gunning Fog Index, SMOG Index) to compare AI-simplified informed consent forms in Plain Language against their originals, finding that prompt strategy significantly affects readability outcomes. Degenhardt (2025) uses UD-based syntactic proxies (e.g. adverbial and relative clause modifiers, passive constructions) to assess syntactic complexity in LLM-generated Easy French translations. Both studies illustrate the range of readability and complexity measures applicable to simplified text, though neither targets German or evaluates conformity to a codified rule set.

The work most closely related to ours is the GermEval 2024 Shared Task on Statement Segmentation in German Easy Language (StaGE; Schomacker et al., 2024). Using BERT-based classifiers, it directly targets one specific Easy Language rule: each sentence should contain exactly one statement. Participating systems outperformed naive baselines, but exact segmentation of statement spans remained challenging (Schomacker et al., 2024). This shows that even a single, well-defined Easy Language rule is non-trivial to automate reliably. To our knowledge, no existing system extends rule-level compliance checking to the full set of Easy or Plain Language requirements, or applies it to both varieties within a single frame-

work.

3 Plain Language vs. Easy Language

German distinguishes between two standardized varieties of simplified language, Plain Language and Easy Language (Maaß, 2020). While both varieties share the overarching goal of making written content more accessible, they differ substantially in target audience, scope, and degree of simplification. Both varieties are governed by official German standards: Easy Language by DIN SPEC 33429 (Deutsches Institut für Normung, 2025), building on earlier guidelines from the Netzwerk Leichte Sprache (Network Easy Language) (Netzwerk Leichte Sprache, 2014), Plain Language by DIN 8581-1 (Deutsches Institut für Normung, 2024).

Concerning the target audience, Easy Language is primarily designed for individuals with cognitive or intellectual disabilities, reading difficulties, or limited literacy. Plain Language, by contrast, addresses a broader audience, including second-language learners, individuals with limited formal education, or readers seeking quick comprehension of complex content such as legal or administrative texts. As a result, Plain Language prioritizes broader acceptability and reach, whereas Easy Language prioritizes maximal comprehensibility for a more narrowly defined target group (Deutsches Institut für Normung, 2024, 2025).

A structural difference is that Plain Language focuses exclusively on textual content, while Easy Language additionally encompasses illustrations and layout as integral components of the standard, for instance mandatory line breaks after each sentence or the use of supporting images. This makes Easy Language a multi-modal accessibility standard rather than a purely linguistic one (Deutsches Institut für Normung, 2025).

Easy Language imposes substantially stricter constraints than Plain Language for the simplification. Easy Language rules prohibit or strongly discourage constructions such as the passive voice, the subjunctive mood, genitive constructions, compound words (unless hyphenated), and metaphorical language, while requiring one statement per sentence and per line. Plain Language relaxes many of these constraints, allowing longer sentences and a wider range of grammatical structures, while still favoring active voice, short sentences, and familiar vocabulary.

Text examples To give a better idea about what Plain and Easy language looks like, Table 1 gives some examples. For example, in the text from the DEplain dataset (Stodden et al., 2023) the sentence

(1) *Chemnitz hatte von 1953 bis 1990 den Namen Karl-Marx-Stadt.*
Chemnitz had from 1953 to 1990 the name Karl-Marx-Stadt.

was simplified to

(2) *Von 1953 bis 1990 hieß Chemnitz Karl-Marx-Stadt.*
From 1953 to 1990 bore.name Chemnitz Karl-Marx-Stadt.

It is not directly obvious why the simplified version is better, i.e. easier to comprehend.

Linguistically, the main difference lies in the choice between a support verb construction and a lexical verb. In *Chemnitz hatte ... den Namen Karl-Marx-Stadt*, the relation is expressed through the light verb *haben* (“to have”) and the noun *Name* (“name”). This is a relatively indirect formulation: the reader must infer that “having the name X” means “being called X”. By contrast, *Chemnitz hieß Karl-Marx-Stadt* uses the verb *heißen* (“to be called”), which directly encodes the naming relation.

However, one could also argue the other way: The construction *hatte den Namen Karl-Marx-Stadt* is more explicit, as “haben” (have) is a very frequent verb and the noun ’Name’ directly signals that the sentence is about a name. The verb “heißen” is a strong verb with an irregular preterite form “hieß” involving an ablaut. The simplified version also places the time adverbial at the beginning, resulting in a non-default word order. In this sense, the simplified sentence may be lexically and morphologically more demanding.

4 Datasets

Madina et al. (2023) provide a survey of datasets and resources. Few German resources explicitly label text as Easy or Plain Language and instead use a generic complexity scale. Moreover, large parts of those datasets are not publicly available, partly for licensing reasons. We thus split our discussion in closed and publicly available datasets.

Dataset	Simplified	Original
ASGC EL	Im Schloss Augustusburg gibt es eine spannende Ausstellung zum Mit-Machen.	Spaß für die ganze Familie: Schloss Augustusburg zeigt Ausstellung zu optischen Täuschungen.
ASGC PL	Eine akute Bronchitis ist eine Erkrankung der Atemwege. Bei einer akuten Bronchitis ist die Schleimhaut in den Bronchien entzündet. Eine Bronchitis kann akut sein. Dann heilt sie oft von selbst. Eine akute Bronchitis dauert etwa zwei Wochen. [...]	Häufig tritt eine akute Bronchitis im Rahmen einer Erkältung auf. Dann kommt es zu Beginn meist zu typischen Erkältungsbeschwerden: Kopf- und Gliederschmerzen, Halsschmerzen, Heiserkeit und Schluckbeschwerden sowie Schnupfen und Abgeschlagenheit. [...] Tritt eine akute Bronchitis im Rahmen einer üblichen Erkältung bei sonst Gesunden auf, heilt sie normalerweise innerhalb von rund zwei Wochen aus. [...]
DEplain	Die Städte Zwickau und Chemnitz liegen im Süden des Bundeslandes Sachsen. Nach Leipzig und Dresden ist Chemnitz mit 240.000 Einwohnern die drittgrößte Stadt Sachsens und Zwickau mit 90.000 Einwohnern die viertgrößte. Von 1953 bis 1990 hieß Chemnitz Karl-Marx-Stadt. [...]	Die Städte Zwickau und Chemnitz liegen im Süden des Bundeslandes Sachsen und bilden zusammen einen Ballungsraum. Die Entfernung zwischen den Stadtzentren beträgt Luftlinie nur etwa 35 Kilometer. Nach Leipzig und Dresden ist Chemnitz mit 240.000 Einwohnern die drittgrößte Stadt Sachsens und Zwickau mit 90.000 Einwohnern die viertgrößte. Chemnitz hatte von 1953 bis 1990 den Namen Karl-Marx-Stadt. [...]
G4A	In den Regeln der NATO fehlen Informationen: • Wie kann man ein Land rauswerfen? • Was passiert, wenn ein Land Kriegt anfängt? Es gibt aber einen anderen Vertrag. Der Vertrag heißt Wiener Vertrags-Rechts-Konvention. Der Vertrag ist aus dem Jahr 1969. Der Vertrag sagt: Ein Land hält die Regeln nicht ein. Dann darf das Land nicht mehr mitmachen. Das Land ist dann nicht mehr Teil von der NATO. Die Länder in der NATO müssen friedlich sein. Die Länder müssen Streit friedlich lösen. Alle Länder müssen diese Regel einhalten.	Der Ausschluss eines Mitglieds ist im NATO-Vertrag nicht vorgesehen. Auch der Fall, dass ein Mitgliedstaat einen Angriffskrieg beginnt, ist in dem Vertrag nicht geregelt. Anwendbar ist in einem solchen Fall Art. 60 Abs. 3b der Wiener Vertragsrechtskonvention (WVRK) vom 23. Mai 1969, wonach generell eine “erhebliche Verletzung” eines Vertrags (hier: des NATO-Vertrags) zum Ausschluss des beschuldigten Vertragspartners führen kann. Dies betrifft im Fall des NATO-Vertrags insbesondere die Pflicht aller Vertragspartner, sich um eine friedliche Streitbeilegung zu bemühen.
Leiko PL	Ärger bei Nato-Treffen Viele wichtige Politiker aus der ganzen Welt sind bei dem Treffen von der Nato gewesen. Zum Beispiel der Präsident von den USA, Donald Trump. Auch Bundes-Kanzlerin Angela Merkel war dabei. Die beiden hatten Streit. Und Donald Trump hat auch sonst Ärger gemacht.	-
Leiko EL	2 Millionen Euro verschwunden In Schleswig-Holstein ist viel Geld verschwunden. Das Geld war in einem Geld·transporter. Ein Geld·transporter ist ein besonderes Auto. Ein Geld·transporter ist besonders gesichert. Ein Geld·transporter hat zum Beispiel sehr dicke Türen. Ein Geld·transporter bringt zum Beispiel Geld zu einer Bank.	-

Table 1: Examples of simplified texts and (where available) parallel original versions. EL = Easy Language, PL = Plain Language. ASGC = Aligned Simple German Corpus, G4A = German4All. Information about the corpora can be found in section 4. A translated version of this table can be found in Table 3 in the Appendix A.

Status	Dataset	# Texts	Domain
Orig	ASGC SL	633	Web
	DEplain Web	30	News
	German4All	150	Wiki
Plain	ASGC PL	168	Web
	DEplain Web	30	News
	Leiko PL	108	News
Easy	ASGC EL	492	Web
	German4All	150	Wiki
	Leiko EL	108	News

Table 2: Overview of public datasets used in our experiments. EL = Easy Language, PL = Plain Language, SL = Standard Language. ASGC = Aligned Simple German Corpus. Information about the corpora can be found in section 4.

4.1 Public Datasets

Table 2 gives an overview of all public datasets that we are aware of. All of them are used in our experiments.

Aligned Simple German Corpus ASGC (Toborek et al., 2023) contains pairs of original websites and their Plain or Easy language counterparts (Toborek et al., 2023). The dataset is available through a web crawler that in our case failed to download some of the sources. In the subsequent step of extracting the text content from the HTML sources, some documents fail to process. We performed a manual analysis of the resulting texts removing empty documents or documents containing non-textual content (e.g. only a list of names, probably the authors of the web article).

DEplain-web DEplain (Stodden et al., 2023) is a parallel German corpus of simplifications into Plain Language. The public part of the corpus contains sentence pairs collected from various web sources. We use the small set of non-restricted Plain texts, but remove the biblical texts.

German4All G4A (Anschütz et al., 2025) is a large-scale dataset for readability-controlled paraphrasing in German, comprising over 25,000 paragraphs extracted from German Wikipedia articles. Each source paragraph is paraphrased into five distinct complexity levels using large language models (GPT-4/Claude). The dataset contains mainly synthetic LLM simplifications that are not in the scope of our study. However, it also contains a small portion of 150 texts manually simplified into Easy Language that we use for our experiments.

Leiko (Jablotschkin and Zinsmeister, 2024) contains both Easy and Plain Language from German news websites. explicitly distinguishing the two varieties via four subcorpora based on source website and rule set. Note that the classification in Easy or Plain language is based on how the news outlet labeled their simplified versions. We decided to keep their classification, although based on manual inspection it might not be justified in all cases.

4.2 Restricted Datasets

Unfortunately, due to licensing or other reasons, many datasets of simplified German are either not available at all or with specific restrictions.

DEplain (Stodden et al., 2023) provides besides the public part, a restricted set of parallel document and sentence pairs, crawled from various sources and from APA (Austria Press Agency) news items. There is also a dedicated **APA Corpus** (Säuberli et al., 2020) from the same source. The **Geasy Corpus** (Hansen-Schirra et al., 2021) contains professional translations into Easy Language. **Korpus Einfaches Deutsch (KED)**[1] contains Plain and Easy Language texts from various web sources that can be queried online, but not downloaded.

4.3 Reference Corpus

As a non-simplified reference corpus, we are using the **Tiger** corpus (Brants et al., 2004) as it is a well-known publicly available corpus. It contains material from different newspaper sections, such as politics, economy, culture, and science reporting (Falenska et al., 2018). As it was not collected with simplification in mind, it provides an unbiased reference for what to expect in standard language. Also many simplified datasets contain news data as well, making them comparable along the dimension of domain (although we acknowledge that full comparability is unachievable).

4.4 Comparability of the datasets

The datasets differ substantially with respect to domain, source, construction method, and degree of parallelism. ASGC and DEplain contain original and simplified counterparts, whereas Leiko provides Plain and Easy Language texts without corresponding original versions. TIGER is used only as an external non-simplified reference corpus and is not parallel to any simplified dataset. German4All contains manually simplified Easy Language texts,

[1] https://korpus-einfaches-deutsch.de

but the broader resource also includes synthetic paraphrases, which we exclude.

Consequently, comparisons in this paper should be understood as corpus-level comparisons rather than controlled comparisons of identical source texts across language varieties. Observed differences may reflect the simplified-language label, but also topic, genre, editorial policy, source selection, or intended audience. We therefore focus on whether broad feature patterns recur across resources, and we avoid attributing differences solely to simplification.

5 Measuring Simplicity

Based on the requirements for Plain and Easy Language described in section 3, we measure simplicity both through continuous complexity features as well as more specific features measuring whether a linguistic construct is present. We use spaCy[2] (Honnibal et al., 2020) for preprocessing and use/adapt linguistic features from the LiFT library.[3]

5.1 Continuous Complexity Features

With regard to complexity, we look at token length, sentence length, vocabulary use, and readability in general.

Token Length We expect the Easy and Plain Language corpora to have shorter tokens on average than their standard language counterpart, as long and compound words should be avoided. We measure the average token length in characters for each text.

Sentence Length We expect the Easy and Plain language corpora to have shorter sentences compared to their standard language counterparts, as subordinate sentences should be avoided. We measure the average sentence length in tokens.

Vocabulary We expect the Easy and Plain Language corpora to include more higher frequency words than their standard language counterpart, as foreign and complex words should be avoided.

We operationalize this through frequency counts as implemented in the wordfreq Python library (Speer, 2022). For each token, we compute the Zipf frequency (i.e. the base-10 logarithm of the frequency count) and round that into the closest integer leading to 7 frequency bins – F7 (very common word) to F1 (very rare word). Some words are not recognized, e.g. foreign, compounds, very uncommon words, or spelling errors (although they are rare in this heavily edited test register). All unrecognized words are put into the out-of-vocabulary (OOV) category. We then compute for each text the proportion of tokens within a category.

Readability Readability formulars are high-level features that combine several indicators (e.g. token length, syllable counts, sentence length) into a single number that represents how readable a text is. We expect Plain and Easy Language Corpora to have lower values compared to non-simplified corpora.

We use the Wiener Sachtextformel (Bamberger and Vanecek, 1984), a set of readability formulas specifically developed for German and therefore better suited to our data than language-independent or English-oriented formulas.

5.2 Specific Features

The continuous complexity features described above are mainly meaningful relative to a point of comparison, e.g. if the average sentence length in a news corpus has a certain value, we expect the simplified version to have a lower value. In the following, we describe a small set of specific features that should be avoided in simplified language. We thus measure 'existence' and count a feature as present if it is found at least once in a text.

With this binary operationalization, the average value across all texts in a corpus directly corresponds to the proportion of texts that contain the respective feature. This allows us to measure how widespread a potentially discouraged construction is across documents, rather than how often it occurs in total. Counting occurrences would answer a different question and could obscure this document-level distribution. For example, the same average number of occurrences could result from one occurrence in every text or from multiple occurrences in only a subset of texts.

Subjunctives In simplified language, subjunctives should be avoided. So instead of *'It is important that the patient be informed about the risks.'* one should write *'The patient must get information about the risks.'* In German, we find a relatively precise subset of subjunctives by looking for the 'Mood' attribute in the output of the morphological analyzer. Subjunctives should be coded as 'Sub'.

[2] https://spacy.io/

[3] https://github.com/zesch/linguistic-features-in-text

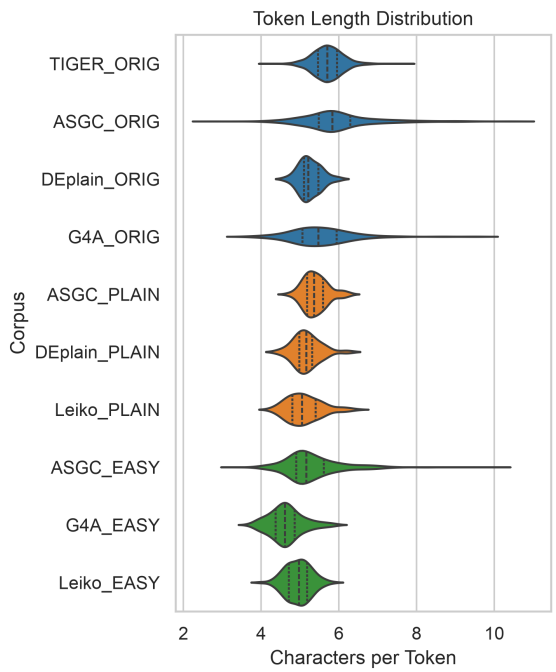


Figure 1: Average token length per corpus.

Note that this is likely to miss other relevant forms of subjunctives, but is still informative for our purposes, as also from this subset there should be near zero instances in ideal simplified language.

Negation In simplified language, unnecessary negation should be avoided. If negation is to be used, only simple negation constructions should be used. As with subjunctives, we only measure a precisely operationalizable subset of single token negation words that should be avoided. We use: *kaum, keinerlei, keinsterlei, keineswegs, keinesfalls, mitnichten, niemand, nichts, niemals, nimmer, nimmermehr, nirgends, nirgendwo, nirgendhin, nirgendher*. We do not include multi-token negations (e.g. *"unter keinen Umständen" (under no circumstances)*), the use of negating prefixes (e.g. *"un·wichtig" (un·important)*) as well as simple negation words that are embedded in a nominal phrase (e.g. *"es gab keinen Hinweis" (there was no clue)*).

6 Results & Discussion

We run all the features on all datasets and report the results here.

Token Length Figure 1 shows the distribution of the average token length per text. While the means of the distributions are not that different (all fall between 4 and 6 characters), the original or reference corpora (with the exception of DEplain) also have longer tokens that are largely absent from the simplified versions. ASGC EL stands out with a higher proportion of long tokens that are mainly compounds like these two examples:

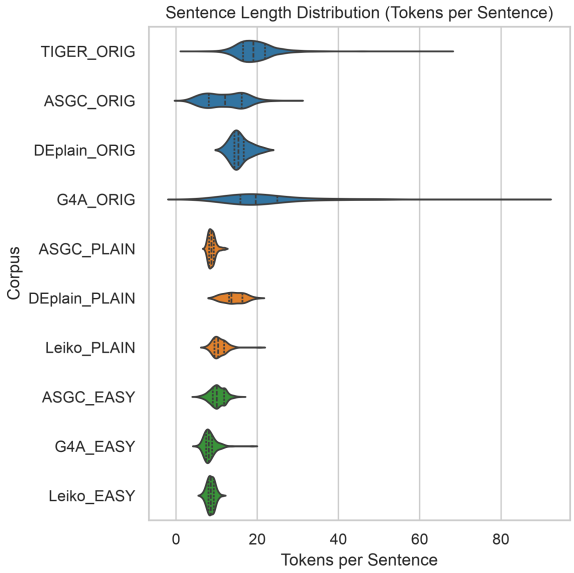


Figure 2: Average sentence length per corpus.

(3) *Bundes-* *Eltern-* *Geld-* *und-* *Eltern-*
Federal- parents- money- and- parents-
Zeit- *Gesetz*
time- law

(4) *Kraft-* *Fahrzeug-* *Haft-Pflicht-*
Power- vehicle- liability-
Versicherung
insurance

Sentence Length Figure 2 shows the distribution of sentence lengths. Here, the expected reduction in sentence length is most clearly visible. As expected, all simplified-language corpora have lower average sentence lengths than the corresponding original or reference corpora. All simplified corpora have sentence lengths below 20 tokens with means around 10 tokens (a bit higher for DEplain).

Vocabulary Figure 3 shows the frequency bins as our proxy for complexity of vocabulary. An effect is mainly visible for F5 and from F3 onward, while F4 is less affected. Table 4 in the Appendix shows the exact values for each corpus.[4] We would

[4]Table 5 in the Appendix additionally shows the same analysis for types instead of tokens, but with similar results.

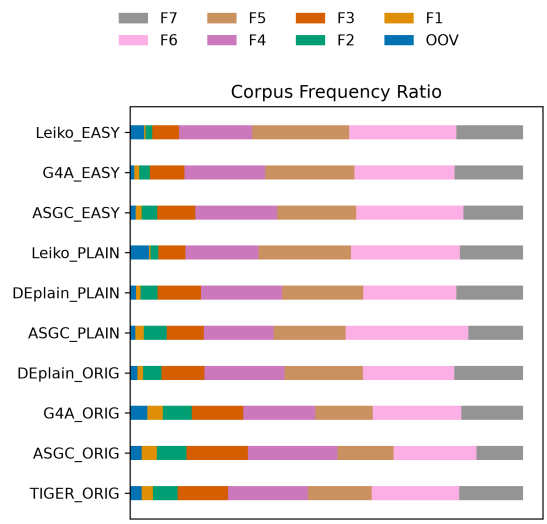


Figure 3: Proportion of tokens with highest corpus frequency (F7) to lowest (F1) as well as out-of-vocabulary (OOV) tokens.

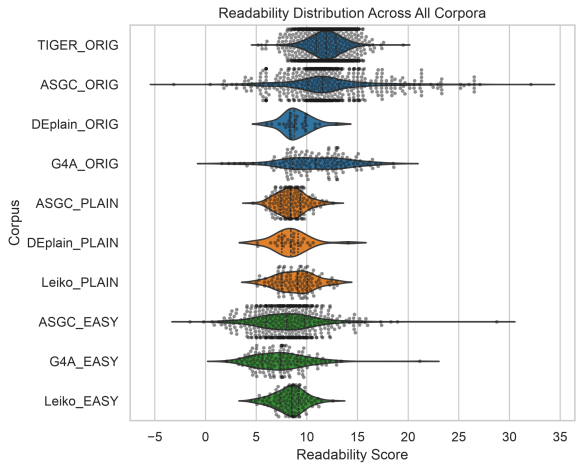


Figure 4: Distribution of readability scores per corpus.

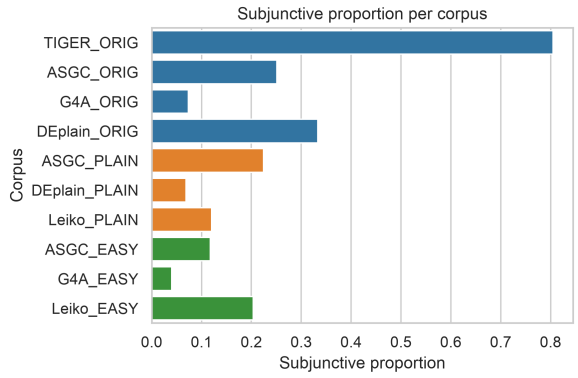


Figure 5: Proportion of documents per corpus containing at least one automatically detected subjunctive. Values represent document-level presence, not the frequency of subjunctive forms within documents.

have expected a stronger impact, but simplification may happen within a band. In addition, domain, corpus construction, and preprocessing artifacts may obscure or mimic differences associated with simplification.

To investigate this further, we inspect the tokens in each frequency band. Starting with OOV, we find occasional spelling errors (e.g. *"Anwäte"* instead of *"Anwälte" (lawyers)*, but mostly numbers (which should be filtered out in improved preprocessing) and compounds. We also find a surprising amount of tokenization (missing whitespace) and encoding issues (probably artifacts from the crawled nature of the data).[5] We conclude that the robustness of the frequency measurement needs to be improved. So far the results can only be considered preliminary.

Readability Figure 4 shows the distribution for the readability scores. This aggregated feature shows a more varied picture, with the ASGC and German4All Easy variants having some examples with quite high readability scores.

Specific Features Figure 5 shows the proportion of documents in each corpus that has at least one **subjunctive**. It is interesting to note that this is true for almost 80% of the documents in the TIGER reference corpus, but is much less prevalent in the original versions of the other corpora. While there is a consistent pattern of the simplified versions having fewer subjunctives than their original counterparts, the differences between corpora in one category are as big as between the original and simplified variants.

The picture is even less clear for **negations** as shown in Figure 6. First, only roughly 20% of documents in the reference corpus contain one of the negations to be avoided, but there are also Plain and Easy variants with around 15% of such negations. Still, the simplified versions have fewer negations than their original counterparts. Further qualitative inspection is needed.

We checked all instances of subjunctives and negations in the Leiko EL corpus (as it contains the largest portion of subjunctives and negations among the Easy language corpora) and found that they are mostly correctly recognized. Thus, although our analysis only covers a subset of possible constructions, it already shows that constructions

[5] In Tables 6 and 7 in the appendix, we show for the Leiko EL corpus the most frequent words in each frequency band.

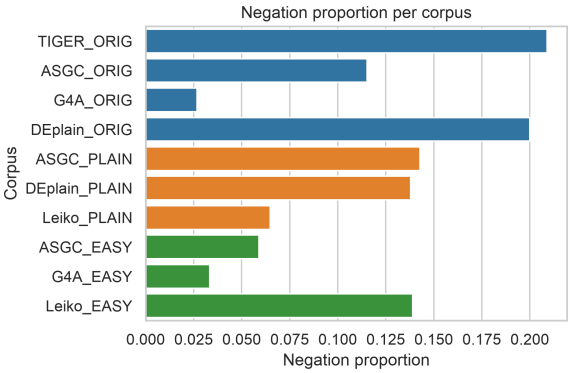


Figure 6: Proportion of documents per corpus containing at least one selected single-token negation form discouraged in simplified language guidelines. Values represent document-level presence, not the frequency of negation forms within documents.

discouraged by simplified language guidelines occur in a substantial proportion of texts.

Examples for subjunctive include:

(5) *Klar, man könnte jetzt mehr Eintritt*
Sure, one could now more admission
nehmen.
take.

(6) *Frauen in Deutschland sollten nicht mehr*
Women in Germany should not more
Politikerinnen sein.
politicians be.

(7) *Frankreich könnte diesmal den Pokal*
France could this time the trophy
gewinnen.
win.

Examples for complex negations include:

(8) *Bei diesem Unfall wurde niemand*
In this accident was nobody
verletzt.
hurt.

(9) *Und es gab kaum Döner- Läden.*
And there was barely Döner- shops.

where *"niemand" (nobody)* in example (8) could be replaced with the simpler *"keine Person" (no person)*, and in example (9) *"kaum" (barely)* could be replaced with the simpler *"sehr wenige" (very few)*.

7 Conclusion & Future Work

This paper asked whether corpora labeled as Plain or Easy Language show the expected reductions in surface-level complexity and guideline-discouraged phenomena, and whether these patterns are stable across datasets. The results partly confirm these expectations: sentence length shows the clearest reduction, whereas token length, vocabulary, readability, subjunctives, and negation show more dataset-specific patterns.

In particular, for subjunctives we observe that differences between corpora within the same category can be comparable to the difference between original and simplified versions, raising the question to what extent guideline-related phenomena are affected by simplification itself versus domain, text and dataset construction. For vocabulary-based features, qualitative inspection suggests that out-of-vocabulary tokens are partly driven by corpus specific artifacts such as numbers, compounds and occasional tokenization issues. This limits how directly these measurements can be interpreted as simplification effects and leaves open how to operationalize the detection of simple vocabulary robustly across heterogeneous sources. Similarly, the mixed picture for negation and readability points to open questions about which automatically measurable indicators align most reliably with simplification goals, and how sensitive these measures are to preprocessing choices.

Finally, our qualitative exemplary analysis suggests that simplification can reduce one type of difficulty while increasing another (e.g. syntactic vs. lexical or morphological difficulty). This raises the open question of how such trade-offs can be identified systematically, for example by analyzing aligned sentence pairs where available, and how they should be evaluated with respect to the intended target audience.

Future work should focus on improving the robustness of the pipeline, especially for vocabulary-based measures, by addressing preprocessing artifacts such as tokenization issues, encoding problems, numbers, and compounds. In addition, we plan to extend the set of specific features toward further guideline-relevant phenomena and to complement the corpus-level analysis with more fine-grained inspection where sentence-level alignment is available.

Limitations

The observed differences cannot be attributed solely to Easy Language or Plain Language, but may also reflect corpus composition and dataset construction.

We use the Easy Language and Plain Language labels provided by the original datasets and do not independently verify full compliance with the corresponding standards. A complete rule-level assessment would require expert annotation and the operationalization of many additional linguistic, layout-related, and multi-modal requirements, which is beyond the scope of this paper.

We focus on a small number of indicators, such as sentence length, token length, lexical frequency, readability, selected subjunctive forms, and selected complex negation expressions. Other relevant phenomena, including passive voice, genitive constructions, tense, compound splitting, and the one-statement-per-sentence principle, are left for future work. Moreover, for the specific guideline-related features, we use a binary presence measure, which indicates whether a phenomenon occurs at all but does not capture its normalized frequency.

Our results also depend on automatic preprocessing which might be negatively affected by features of simplified language like unusual line breaks, hyphenated compounds, or non-standard punctuation. This is particularly relevant for the lexical-frequency analysis, where low-frequency and out-of-vocabulary tokens may include names, numbers, compounds, domain-specific terms, or preprocessing artifacts rather than genuinely difficult vocabulary.

Finally, the TIGER corpus serves only as an approximate non-simplified reference corpus and is not fully matched to the simplified datasets in topic, genre, or publication context.

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Lay Summary

In this article, we examine texts in German. We compare standard language with two types of simplified language: Easy Language and Plain Language. We treat both as separate targets. We use datasets that are publicly available. We measure various features that indicate how difficult a text is. These include features related to text length and grammar, such as sentence structure. We compare each original text with its simplified version. Then we examine whether Easy Language and Plain Language are similar across all datasets, or whether they differ significantly depending on the corpus. In this way, we show how simplified language is actually expressed in different collections of texts.

German version In diesem Artikel untersuchen wir Texte auf Deutsch. Wir vergleichen Standardsprache mit zwei Arten von vereinfachter Sprache: Leichte Sprache und Einfache Sprache. Wir behandeln beide als eigene Ziele. Wir nutzen Datensätze, die öffentlich verfügbar sind. Wir messen verschiedene Merkmale, die zeigen, wie schwierig ein Text ist. Dazu gehören Merkmale zur Textlänge und zur Grammatik (zum Beispiel Satzbau). Wir vergleichen jeweils den Originaltext mit der vereinfachten Version. Dann prüfen wir: Sind Leichte Sprache und Einfache Sprache in allen Datensätzen ähnlich? Oder unterscheiden sie sich je nach Korpus deutlich? So zeigen wir, wie sich vereinfachte Sprache in verschiedenen Sammlungen von Texten tatsächlich ausdrückt.

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A Translated Data

Dataset	Simplified	Original
ASGC EL	Augustusburg Palace is hosting an exciting hands-on exhibition.	Fun for the whole family: Augustusburg Palace is hosting an exhibition on optical illusions.
ASGC PL	Acute bronchitis is a respiratory tract condition. In acute bronchitis, the lining of the bronchi becomes inflamed. Bronchitis can be acute. In such cases, it often clears up on its own. Acute bronchitis lasts about two weeks. [...]	Acute bronchitis often occurs as part of a cold. In such cases, it usually begins with typical cold symptoms: headache, body aches, sore throat, hoarseness, and difficulty swallowing, as well as a runny nose and fatigue. [...] If acute bronchitis occurs as part of a common cold in otherwise healthy individuals, it usually resolves within about two weeks. [...]
DEplain	The cities of Zwickau and Chemnitz are located in the southern part of the state of Saxony. After Leipzig and Dresden, Chemnitz—with a population of 240,000—is Saxony's third-largest city, and Zwickau—with a population of 90,000—is the fourth-largest. From 1953 to 1990, Chemnitz was known as Karl-Marx-Stadt. [...]	The cities of Zwickau and Chemnitz are located in the southern part of the state of Saxony and together form a metropolitan area. The distance between the city centers is only about 35 kilometers as the crow flies. After Leipzig and Dresden, Chemnitz, with 240,000 residents, is Saxony's third-largest city, and Zwickau, with 90,000 residents, is the fourth-largest. From 1953 to 1990, Chemnitz was known as Karl-Marx-Stadt. [...]
G4A	There is information missing from NATO's rules: • How can you kick a country out? • What happens when a country starts a war? However, there is another treaty. The treaty is called the Vienna Convention on the Law of Treaties. The treaty dates from 1969. The treaty states: If a country does not comply with the rules, then that country may no longer participate. That country is then no longer a member of NATO. NATO member countries must be peaceful. Countries must resolve disputes peacefully. All countries must abide by this rule.	The NATO Treaty does not provide for the expulsion of a member. Nor does the Treaty address the situation in which a member state initiates a war of aggression. In such a case, Article 60, paragraph 3b, of the Vienna Convention on the Law of Treaties (VCLT) of May 23, 1969, according to which a "material breach" of a treaty (in this case, the NATO Treaty) can generally lead to the expulsion of the accused contracting party. In the context of the NATO Treaty, this applies in particular to the obligation of all contracting parties to strive for the peaceful settlement of disputes.
Leiko PL	Trouble at the NATO Summit Many important politicians from around the world attended the NATO summit. For example, U.S. President Donald Trump. German Chancellor Angela Merkel was also there. The two had a disagreement. And Donald Trump caused trouble in other ways as well.	-
Leiko EL	2 million euros gone A lot of money has gone missing in Schleswig-Holstein. The money was in an armored truck. An armored truck is a special kind of vehicle. An armored truck is specially secured. For example, an armored truck has very thick doors. For example, an armored truck transports money to a bank.	-

Table 3: Examples of simplified texts and (where available) parallel original versions. EL = Easy Language, PL = Plain Language. ASGC = Aligned Simple German Corpus, G4A = German4All. Information about the corpora can be found in section 4. The original German version of this table can be found in Table 1.

B Detailed Results

Variant	Corpus	F7	F6	F5	F4	F3	F2	F1	OOV
Ref	TIGER	.16	.22	.16	.20	.13	.06	.03	.03
Orig	ASGC	.12	.21	.14	.23	.16	.08	.04	.03
	G4A	.16	.23	.15	.18	.13	.07	.04	.04
	DEplain	.18	.23	.20	.20	.11	.05	.02	.02
Plain	ASGC PL	.14	.31	.18	.18	.10	.06	.02	.01
	DEplain	.17	.24	.21	.21	.11	.04	.01	.02
	Leiko PL	.16	.28	.24	.19	.07	.02	.00	.05
Easy	ASGC EL	.15	.27	.20	.21	.10	.04	.02	.01
	G4A	.18	.26	.23	.21	.09	.03	.01	.01
	Leiko EL	.17	.27	.25	.19	.07	.02	.00	.04

Table 4: Proportion of tokens within frequency band per corpus.

Variant	Corpus	F7	F6	F5	F4	F3	F2	F1	OOV
Ref	TIGER	.05	.18	.19	.26	.16	.08	.04	.04
Orig	ASGC	.06	.16	.16	.27	.18	.09	.04	.04
	G4A	.11	.22	.16	.20	.15	.08	.04	.04
	DEplain	.05	.19	.24	.27	.15	.07	.02	.02
Plain	ASGC PL	.04	.21	.26	.26	.12	.06	.02	.02
	DEplain	.07	.22	.24	.25	.13	.06	.02	.02
	Leiko PL	.09	.26	.28	.21	.08	.02	.00	.05
Easy	ASGC EL	.12	.24	.22	.23	.11	.04	.02	.02
	G4A	.12	.27	.25	.22	.09	.03	.01	.01
	Leiko EL	.07	.26	.31	.23	.08	.02	.00	.04

Table 5: Proportion of types within frequency band per corpus.

F7		**F6**		**F5**		**F4**	
die	1306	hat	441	menschen	454	mannschaft	82
in	766	von	433	viele	297	flüchtlinge	62
der	739	sie	414	deshalb	219	politiker	56
und	656	ein	377	beispiel	208	feuer	52
ist	635	eine	316	deutschland	137	tor	44
das	549	es	305	gesagt	108	behinderung	37
den	316	sind	288	oft	90	gericht	32
nicht	279	zum	281	frauen	84	parteien	31
zu	166	für	277	polizei	83	wm	31
ich	109	haben	268	hamburg	82	niedersachsen	30

Table 6: Top-10 most frequent tokens in the Leiko PL corpus for the four highest frequency bands (F7–F4).

F3		**F2**		**F1**		**OOV**	
behinderungen	64	carles	25	kappert-gonther	11	fach·leute	22
taz	64	puigdemont	25	tschentscher	11	nord·bank	19
pille	25	hsh	19	langensiepen	10	bürger·meister	17
keime	24	büchereien	17	verhütungs-mittel	8	nord·deutschland	16
demonstrationen	22	migrations-hintergrund	17	buntenbach	5	feier·tag	13
fähre	20	kul	15	219a	4	geld·transporter	13
nachspielzeit	16	baris	14	birkel	4	polizei·schule	13
südkorea	16	freibäder	14	friesacher	4	segel·schiff	13
abkürzung	15	koalitions-vertrag	13	schlagermove	4	bundes·garten·schau	12
scholz	15	familien-nachzug	10	guha	3	kreuz·fahrt·schiff	12

Table 7: Top-10 most frequent tokens in the Leiko PL corpus for the lower frequency bands (F3–OOV).